

18B.15. Diffusion and heterogeneous reaction in a slender cylindrical tube with a closed end (Fig. 18B.15). A slender cylindrical pore of length L , cross-sectional area S , and perimeter P , is in contact at its open end with a large body of well-mixed fluid, consisting of species A and B . Species A , a minor constituent of this fluid, disappears into the pore, diffuses in the z direction and reacts on its walls. The rate of this reaction may be expressed as $(\mathbf{n} \cdot \mathbf{n}_s)|_{\text{surface}} = f(\omega_{A0})$; that is, at the wall the mass flux normal to the surface is some function of the mass fraction, ω_{A0} , of A in the fluid adjacent to the solid surface. The mass fraction ω_{A0} depends on z , the distance from the inlet. Because A is present in low concentration, the fluid temperature and density may be considered constant, and the diffusion flux is adequately described by $\mathbf{j}_A = -D_{AB} \nabla \omega_A$.

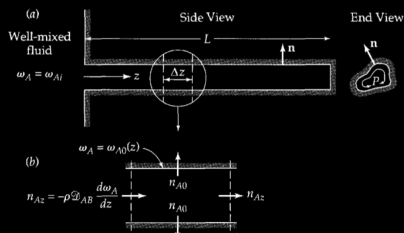


Fig. 18B.15. (a) Diffusion and heterogeneous reaction in a long, non-circular cylinder. (b) Region of thickness Δz over which the mass

Concentration Distributions in Solids and in Laminar Flow

where the diffusivity may be regarded as a constant. Because the pore is long compared to its lateral dimension, concentration gradients in the lateral directions may be neglected. Note the similarity with the problem discussed in §10.7.

(a) Show by means of a shell balance that, at steady state,

$$\frac{dn_{Az}}{dz} = \frac{P}{S} f(\omega_{A0}) \quad (18B.15-1)$$

(b) Show that the steady-state mass average velocity v_z is zero for this system.

(c) Substitute the appropriate form of Fick's law into Eq. 18.15-1, and integrate the resulting differential equation for the special case that $f(\omega_{A0}) = K' \omega_{A0}$. To obtain a boundary condition at $z = L$, neglect the rate of reaction on the closed end of the cylinder; why is this a reasonable approximation?

(d) Develop an expression for the total rate w_A of disappearance of A in the cylinder.

(e) Compare the results of parts (c) and (d) with those of §10.7 both from the standpoint of the mathematical development and the nature of the assumptions made.

Answers: (c) $\frac{\omega_A}{\omega_{Ai}} = \frac{\cosh N[1 - (z/L)]}{\cosh N}$, where $N = \sqrt{\frac{PL^2 K'}{S D_{AB}}}$; (d) $w_A = (S P D_{AB} \omega_{Ai} / L) N \tanh N$

$$d.) \quad W_A = S \cdot n_{Az}|_{z=0} = -S g D_{AB} \left. \frac{d\omega_A}{dz} \right|_{z=0}$$

$$\frac{d}{dz} (W_A) = \frac{d}{dz} \left(W_{Ai} \frac{\cosh N[1 - \frac{z}{L}]}{\cosh N} \right) = W_{Ai} \frac{-\frac{N}{L} \sinh N \left(1 - \frac{z}{L} \right)}{\cosh N} \Big|_{z=0}$$

$$\frac{dW_A}{dz} = -W_{Ai} \frac{N}{L} \tanh(N)$$

$$W_A = -S g D_{AB} \left(-W_{Ai} \frac{N}{L} \tanh(N) \right) =$$

$$W_A = \left(\frac{S g D_{AB} W_{Ai}}{L} \right) \cdot N \tanh(N)$$



diffusion at diffusion at $z+\Delta z$ Loss of $P \Delta z$ in outward direction

$$a.) \quad \lim_{\Delta z \rightarrow 0} \frac{[S n_{Az}]_z - [S n_{Az}]_{z+\Delta z} - P \Delta z f(\omega_{A0})}{P \Delta z} = 0$$

$$-\frac{dn_{Az}}{dz} \left(\frac{S}{P} \right) - f(\omega_{A0}) = 0$$

$$-\frac{dn_{Az}}{dz} = \frac{P}{S} f(\omega_{A0})$$

$$b.) \quad v_{z,avg} = \frac{1}{S} \int_0^L v_z(z) dz = \frac{0}{S} = 0$$

$v_z(z) = 0$, $\frac{dv_z}{dz} = 0$ only diffusion w/o velocity exists.

$$c.) \quad -\frac{d}{dz} (n_{Az}) = \frac{P}{S} K' \omega_{A0} \quad | \quad n_{Az} = -P D_{AB} \frac{d\omega_A}{dz}$$

$$\frac{d}{dz} \left(g D_{AB} \frac{d\omega_A}{dz} \right) = \frac{P}{S} K' \omega_{A0}$$

$$\frac{d^2 \omega_A}{dz^2} - \frac{P K'}{S g D_{AB}} \omega_{A0}(z) = 0$$

$$BC: \quad \omega_A(z=0) = W_{Ai} \quad \left. \frac{d\omega_A}{dz} \right|_{z=L} = 0 \text{ (finite)}$$

$$\lim_{z \rightarrow 0} \omega_{A0}(z) = W_A$$

$$\frac{d^2 W_A}{dz^2} - \frac{P K'}{S g D_{AB}} W_A = 0$$

$$y'' - m^2 y = 0 \quad r^2 - \left(\frac{N}{L} \right)^2 \rightarrow r = \pm \frac{N}{L}$$

$$\frac{d^2 W_A}{dz^2} - \left(\frac{P K' L^2}{S g D_{AB}} \right) W_A = 0$$

$$\frac{d^2 W_A}{dz^2} - \frac{N^2}{L^2} W_A = 0$$

$$W_A(z) = C_1 \cosh\left(\frac{N}{L} z\right) + C_2 \sinh\left(\frac{N}{L} z\right)$$

$$W_{Ai} = C_1 \cosh(0) + C_2 \sinh(0)$$

$$W_{Ai} = C_1$$

$$W_A(z) = W_{Ai} \cosh\left(\frac{N}{L} z\right) + C_2 \sinh\left(\frac{N}{L} z\right)$$

$$\left. \frac{dW_A}{dz} \right|_{z=L} = 0 = W_{Ai} \frac{N}{L} \sinh(N) + C_2 \cosh\left(\frac{N}{L} L\right)$$

$$C_2 = -W_{Ai} \frac{\sinh(N)}{\cosh(N)} = -W_{Ai} \tanh(N)$$

$$W_A = W_{Ai} \cosh\left(\frac{N}{L} z\right) - W_{Ai} \tanh(N) \sinh\left(\frac{N}{L} z\right)$$

$$W_A = \frac{W_{Ai}}{\cosh N} \left[\cosh\left(\frac{N}{L} z\right) \cosh(N) - \sinh(N) \sinh\left(\frac{N}{L} z\right) \right]$$

$$\cosh(A-B) \quad A=N, B=\frac{Nz}{L}$$

$$W_A = \frac{W_{Ai}}{\cosh N} \cosh\left(N - \frac{Nz}{L}\right)$$

$$\frac{W_A}{W_{Ai}} = \frac{\cosh\left(N\left[1 - \frac{z}{L}\right]\right)}{\cosh(N)}$$

reasonable assuming diffusion through the membrane is significantly greater than the diffusion in z -direction.

18C.3. Absorption rate in a falling film. The result in Eq. 18.5-18 may be obtained by an alternative procedure.

(a) According to an overall mass balance on the film, the total moles of A transferred per unit time across the gas-liquid interface must be the same as the total molar rate of flow of A across the plane $z = L$. The latter rate is calculated as follows:

$$W_A = \lim_{\delta \rightarrow \infty} (W \delta v_{\max}) \left(\frac{1}{\delta} \int_0^{\delta} c_A|_{z=L} dx \right) = W v_{\max} \int_0^{\infty} c_A|_{z=L} dx \quad (18C.3-1)$$

Explain this procedure carefully.

(b) Insert the solution for c_A in Eq. 18.5-15 into the result of (a) to obtain:

$$W_A = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \int_0^{\infty} \left(\int_0^{\infty} \frac{x}{x + \sqrt{4 D_{AB} z} / v_{\max}} \exp(-\xi^2) d\xi \right) dx \\ = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \sqrt{\frac{4 D_{AB} L}{v_{\max}}} \int_0^{\infty} \left(\int_0^{\infty} \exp(-\xi^2) d\xi \right) du \quad (18C.3-2)$$

In the second line, the new variable $u = x / \sqrt{4 D_{AB} z} / v_{\max}$ has been introduced.

(c) Change the order of integration in the double integral, to get

$$W_A = W L c_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}} \cdot 2 \int_0^{\infty} \exp(-\xi^2) \left(\int_0^{\xi} du \right) d\xi \quad (18C.3-3)$$

Explain by means of a carefully drawn sketch how the limits are chosen for the integrals. The integrals may now be done analytically to get Eq. 18.5-18.

$$W_A = \int_0^W \int_0^L N_{Ax}|_{x=0} dz dy \\ = W c_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi}} \int_0^L \frac{1}{\sqrt{z}} dz \\ = W L c_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}} \quad (18.5-18)$$

$$v_z(x) = v_{\max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right]$$

$$\frac{[W \delta x N_{Az}]_z - [W \delta x N_{Az}]_{z+\Delta z} + [W \delta z N_{Ax}]_x - [W \delta z N_{Ax}]_{x+\Delta x}}{W \delta x \delta z} = 0$$

$$\frac{\partial N_{Az}}{\partial z} + \frac{\partial N_{Ax}}{\partial x} = 0$$

$$-D_{AB} \frac{\partial c_A}{\partial z} + x_A (N_{Az} + N_{Bz}) \approx c_A v_z(x) \leftarrow \text{diffusion of A primarily due to Convection} \gg j_A$$

$$\frac{\partial N_{Ax}}{\partial x} \approx -D_{AB} \frac{\partial c_A}{\partial x}$$

$$v_z(x) \frac{\partial c_A}{\partial z} = D_{AB} \frac{\partial^2 c_A}{\partial x^2}$$

$$v_{\max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \frac{\partial c_A}{\partial z} = D_{AB} \frac{\partial^2 c_A}{\partial x^2}$$

$$N_{Ax}|_{x=0} = c_{A0} \sqrt{\frac{D_{AB} v_{\max}}{\pi z}}$$

BC:

$$c_A(z=0, x) = 0 \text{ Pure B @ top}$$

$$c_A(z, x=0) = c_{A0} \text{ Diffusion of A} \rightarrow \text{B}$$

$$\frac{\partial c_A}{\partial x} \Big|_{x=\delta} = 0 \text{ Non-diffusible surface} \rightarrow c_A(z, x=\infty) = 0 \text{ for short instance}$$

$$W_A = \lim_{\delta \rightarrow \infty} (W \delta v_{\max}) \left(\frac{1}{\delta} \int_0^{\delta} c_A|_{z=L} dx \right)$$

$$W_A = \int_0^{\infty} c_A|_{z=L} dx$$

lim $\delta \rightarrow \infty$ = film is short or diffusing slowly A never truly meets solid wall at $x = \delta$

$v_{\max}$ = A remains primarily near $x=0$ experiencing max velocity where $c_A > 0$

$\frac{1}{\delta} \int_0^{\delta} c_A|_{z=L} dx$ = Average concentration across the film which when multiplied by

$W \delta v_{\max}$ gives W_A .

$$b.) \frac{c_A}{c_{A0}} = 1 - \frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{4 D_{AB} z} / v_{\max}} \exp(-\xi^2) d\xi \Rightarrow c_A = c_{A0} \operatorname{erfc} \left(\frac{x}{\sqrt{4 D_{AB} z} / v_{\max}} \right)$$

$$W_A = \int_0^{\delta} \left(c_{A0} \left(1 - \frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{4 D_{AB} z} / v_{\max}} \exp(-\xi^2) d\xi \right) \right) dx$$

$$\xrightarrow[\text{Wolfram Alpha}]{\text{Some Bullshit}} W_A = W v_{\max} c_{A0} \frac{2}{\sqrt{\pi}} \sqrt{\frac{4 D_{AB} L}{v_{\max}}} \int_0^{\infty} \left(\int_u^{\infty} \exp(-\xi^2) d\xi \right) du$$

$$c.) \xrightarrow[\text{Wolfram Alpha}]{\text{More bullshit}} W_A = W L c_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}} \cdot 2 \int_0^{\infty} \exp(-\xi^2) \left(\int_0^{\xi} du \right) d\xi$$

Original BC:

$$0 \leq u \leq \infty, 0 \leq \xi \leq \infty$$

New Limits:

$$0 \leq \xi \leq \infty$$

$$0 \leq u \leq \xi$$

$$\int_0^{\infty} \left(\int_u^{\infty} \exp(-\xi^2) d\xi \right) du \\ = \int_0^{\infty} \exp(-\xi^2) \left(\int_0^{\xi} du \right) d\xi = \int_0^{\infty} \xi \exp(-\xi^2) d\xi$$

$$s = \xi^2 \quad ds = 2\xi d\xi$$

$$\frac{1}{2} \int_0^{\infty} \exp(-s) ds = \frac{1}{2} \left[\exp(-s) \right]_0^{\infty} = \frac{1}{2} (0 + 1) = \frac{1}{2}$$

$$W_A = W L c_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}}$$

19B.1. **Steady-state evaporation** (Fig. 18.2-1). Rework the problem solved in §18.2, dealing with the evaporation of liquid A into gas B, starting from Eq. 19.1-17.

(a) First obtain an expression for v^* , using Eq. (M) of Table 17.8-1, as well as Fick's law in the form of Eq. (D) of Table 17.8-2.

(b) Show that Eq. 19.1-17 then becomes the following nonlinear second-order differential equation:

$$\frac{d^2 x_A}{dz^2} + \frac{1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 = 0 \quad (19B.1-1)$$

(c) Solve this equation to get the mole fraction profile given in Eq. 18.2-11.

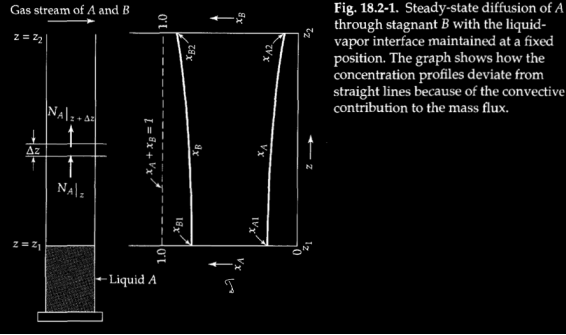


Fig. 18.2-1. Steady-state diffusion of A through stagnant B with the liquid-vapor interface maintained at a fixed position. The graph shows how the concentration profiles deviate from straight lines because of the convective contribution to the mass flux.

a.)

$$c \left(\frac{\partial x_A}{\partial t} + (\mathbf{v}^* \cdot \nabla x_A) \right) = c D_{AB} \nabla^2 x_A + (x_B R_A - x_A R_B) \quad \text{Eq. 19.1-17}$$

$$v^* = \frac{D_{AB} \nabla^2 x_A + \frac{(x_B R_A - x_A R_B)}{c}}{\nabla x_A} \Rightarrow v^* = D_{AB} \nabla x_A + \frac{(x_B R_A - x_A R_B)}{c \nabla x_A}$$

$$N_A = x_A(N_A + N_B) - c D_{AB} \nabla x_A = c_A v^* - c D_{AB} \nabla x_A \Rightarrow v^* = \frac{N_A + c D_{AB} \nabla x_A}{c_A}$$

$$\sum_{\alpha=1}^N N_{\alpha} = c v^* \Rightarrow v^* = \frac{\sum_{\alpha=1}^N N_{\alpha}}{c}$$

b.)

$$c \left(\frac{\partial x_A}{\partial t} + (\mathbf{v}^* \cdot \nabla x_A) \right) = c D_{AB} \nabla^2 x_A + (x_B R_A - x_A R_B) \quad \text{No rate of production of A or B}$$

$$c v^* \frac{dx_A}{dz} = c D_{AB} \frac{d^2 x_A}{dz^2} + (x_B R_A - x_A R_B)$$

$$c v^* \frac{dx_A}{dz} = c D_{AB} \frac{d^2 x_A}{dz^2}$$

$$c v^* = \sum_{\alpha=1}^N N_{\alpha} = N_A + N_B$$

$$N_{A,z} = x_A(N_A + N_B) - c D_{AB} \frac{dx_A}{dz}$$

$$N_{A,z} = -\frac{c D_{AB}}{1-x_A} \frac{dx_A}{dz}$$

$$\left(-\frac{c D_{AB}}{1-x_A} \frac{dx_A}{dz} \right) \frac{dx_A}{dz} - c D_{AB} \frac{d^2 x_A}{dz^2} = 0$$

$$\frac{1}{1-x_A} \left(\frac{dx_A}{dz} \right)^2 + \frac{d^2 x_A}{dz^2} = 0$$

$$u = \frac{dx_A}{dz}, \quad \frac{d^2 x_A}{dz^2} \cdot \frac{du}{dz} = \frac{du}{dx_A} \cdot \frac{dx_A}{dz} \rightarrow u \frac{du}{dx_A}$$

$$\frac{u^2}{1-x_A} + u \frac{du}{dx_A} = 0$$

$$\frac{u}{1-x_A} = -\frac{du}{dx_A}$$

$$-\frac{dx_A}{1-x_A} = \frac{du}{u}$$

$$\ln(C_1) + \ln(1-x_A) = \ln(u)$$

$$C_1(1-x_A) = \frac{dx_A}{dz}$$

$$C_1 dz = \frac{dx_A}{1-x_A}$$

$$C_2 + z C_1 = \ln(1-x_A)$$

$$1-x_A = \exp(C_2 + C_1 z)$$

$$1-x_{A1} = \exp(C_1 z_1 + C_2)$$

$$1-x_{A2} = \exp(C_1 z_2 + C_2)$$

$$\frac{1-x_{A1}}{1-x_{A2}} = \exp(C_1(z_1 - z_2))$$

$$C_1 = \ln \left(\frac{1-x_{A1}}{1-x_{A2}} \right) \cdot \frac{1}{z_1 - z_2}$$

$$\frac{1-x_{A1}}{1-x_{A2}} = \exp \left(\ln \left(\frac{1-x_{A1}}{1-x_{A2}} \right) \cdot \frac{1}{z_1 - z_2} (z - z_2) \right)$$

$$\Gamma = C_1 \cosh(\Phi \xi) + C_2 \sinh(\Phi \xi)$$

$$1 = C_1 \cosh(\Phi) + C_2 \sinh(\Phi)$$

$$1 = C_1$$

$$\Gamma = \cosh(\Phi \xi) + C_2 \sinh(\Phi \xi)$$

$$\frac{\partial \Gamma}{\partial \xi} = 0 = \sinh(\Phi) + C_2 \cosh(\Phi)$$

$$C_2 = -\frac{\sinh(\Phi)}{\cosh(\Phi)} = -\tanh(\Phi)$$

$$\Gamma = \cosh(\Phi \xi) - \tanh(\Phi) \sinh(\Phi \xi)$$

$$= \frac{\cosh(\Phi \xi) \cosh(\Phi) - \sinh(\Phi) \sinh(\Phi \xi)}{\cosh(\Phi)}$$

$$\Gamma = \frac{\cosh[\Phi(1-\xi)]}{\cosh(\Phi)} = \frac{C_A}{C_{A0}} = \frac{\cosh[\Phi(1-\frac{z}{L})]}{\cosh(\Phi)}$$

19B.2. **Gas absorption with chemical reaction** (Fig. 18.4-1). Rework the problem solved in §18.4, by starting with Eq. 19.1-16. What assumptions do you have to make in order to get Eq. 18.4-4?

$$D_{AB} \frac{d^2 c_A}{dz^2} - k_1' c_A = 0 \quad (18.4-4)$$

$$\rho \left(\frac{\partial \omega_A}{\partial t} + (\mathbf{v} \cdot \nabla \omega_A) \right) = \rho D_{AB} \nabla^2 \omega_A + r_A \quad (19.1-16)$$

$$g \left(\frac{\partial \omega_A}{\partial t} + (\mathbf{v} \cdot \nabla \omega_A) \right) = g D_{AB} \nabla^2 \omega_A + r_A$$

Steady State
and mass is relatively close to mole

$$0 = g D_{AB} \nabla^2 \omega_A + r_A$$

- Assumptions:
- @ S.S
 - $\Delta T = 0$
 - $\Delta j = 0$
 - $g \approx M_{wA}$ are constant
 - no convection
 - $C_A \ll 1$
 - One dimensional problem

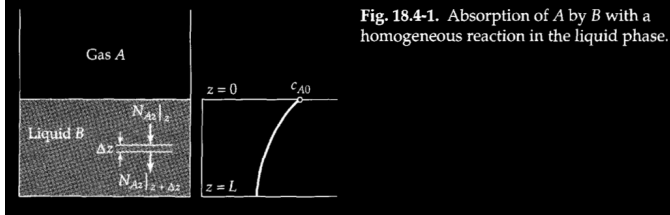


Fig. 18.4-1. Absorption of A by B with a homogeneous reaction in the liquid phase.

$$0 = D_{AB} \frac{d^2 c_A}{dz^2} + r_A \quad r_A = M_A R_A$$

$$0 = D_{AB} M_A \frac{d^2 c_A}{dz^2} + r_A$$

$$0 = D_{AB} \frac{d^2 c_A}{dz^2} + \frac{r_A}{M_A} = D_{AB} \frac{d^2 c_A}{dz^2} + R_A$$

$$0 = D_{AB} \frac{d^2 c_A}{dz^2} - K_1' c_A$$

$$\frac{K_1' L^2}{D_{AB}} c_A = \frac{d^2 c_A}{dz^2} \quad \text{BC: } c_A(z=0) = C_{A0}$$

$$\frac{d c_A}{dz} \Big|_{z=L} = 0 \quad \leftarrow \text{finite } c_A$$

$$\frac{K_1' L^2}{D_{AB}} \Gamma = \frac{d^2 \Gamma}{dz^2}$$

$$\frac{K_1' L^2}{D_{AB}} \Gamma = \frac{1}{L^2} \frac{d^2 \Gamma}{d \xi^2}$$

$$\Phi^2 \Gamma = \frac{d^2 \Gamma}{d \xi^2}$$

$$\xi = \frac{z}{L}$$

$$\Phi = \sqrt{\frac{K_1' L^2}{D_{AB}}}$$

$$\Gamma = \frac{c_A}{C_{A0}}$$

$$\Gamma = \Gamma(c_A(\xi(z)))$$

$$\Gamma(\xi=0) = 1$$

$$\frac{\partial \Gamma}{\partial \xi} \Big|_{\xi=0} = 0$$

$$\Gamma = C_1 \cosh(\Phi \xi) + C_2 \sinh(\Phi \xi)$$

$$1 = C_1 \cosh(\Phi) + C_2 \sinh(\Phi)$$

$$1 = C_1$$

$$\Gamma = \cosh(\Phi \xi) + C_2 \sinh(\Phi \xi)$$

$$\frac{\partial \Gamma}{\partial \xi} = 0 = \sinh(\Phi) + C_2 \cosh(\Phi)$$

$$C_2 = -\frac{\sinh(\Phi)}{\cosh(\Phi)} = -\tanh(\Phi)$$

$$\Gamma = \cosh(\Phi \xi) - \tanh(\Phi) \sinh(\Phi \xi)$$

$$= \frac{\cosh(\Phi \xi) \cosh(\Phi) - \sinh(\Phi) \sinh(\Phi \xi)}{\cosh(\Phi)}$$

$$\Gamma = \frac{\cosh[\Phi(1-\xi)]}{\cosh(\Phi)} = \frac{C_A}{C_{A0}} = \frac{\cosh[\Phi(1-\frac{z}{L})]}{\cosh(\Phi)}$$

19B.4. Oxidation of silicon (Fig. 19B.4).¹ A slab of silicon is exposed to gaseous oxygen (species A) at pressure p , producing a layer of silicon dioxide (species B). The layer extends from the surface $z = 0$, where the oxygen dissolves with concentration $c_{A0} = Kp$, to the surface at $z = \delta(t)$, where the oxygen and silicon undergo a first-order reaction with rate coefficient k_1' . The thickness $\delta(t)$ of the growing oxide layer is to be predicted. A quasi-steady-state method is useful here, inasmuch as the advancement of the reaction front is very slow.

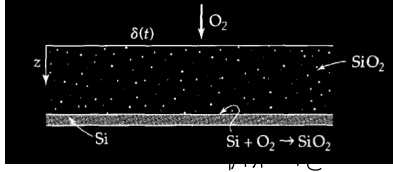


Fig. 19B.4. Oxidation of silicon.

(a) First solve the diffusion equation of Eq. 19.1-18, with the term $\partial c_A / \partial t$ neglected, and apply the boundary conditions to obtain

$$c_A = c_{A0} - (c_{A0} - c_{A\delta}) \frac{z}{\delta} \quad (19B.4-1)$$

in which the concentration $c_{A\delta}$ at the reaction plane is as yet unknown.

(b) Next use an unsteady-state molar O_2 balance on the region $0 < z < \delta(t)$ to obtain, with the aid of the Leibniz formula of §C.3,

$$c_{A\delta} \frac{d\delta}{dt} = -\mathcal{Q}_{AB} \frac{dc_A}{dz} - k_1' c_{A\delta} \quad (19B.4-2)$$

(c) Now write an unsteady-state molar balance on SiO_2 in the same region to obtain

$$+k_1' c_{A\delta} = \frac{1}{V_B} \frac{d\delta}{dt} \quad (19B.4-3)$$

(d) In Eq. 19B.4-2, evaluate $d\delta/dt$ from Eq. 19B.4-3 and dc_A/dz from Eq. 19B.4-1. This will yield an equation for $c_{A\delta}$:

$$\frac{k_1' \delta \bar{V}_B}{\mathcal{Q}_{AB}} c_{A\delta}^2 + \left(1 + \frac{k_1' \delta}{\mathcal{Q}_{AB}}\right) c_{A\delta} = c_{A0} \quad (19B.4-4)$$

Inserting numerical values into Eq. 19B.4-4 shows that the quadratic term can safely be neglected.

(e) Combine Eqs. 19B.4-3 and 19B.4-4 (without the quadratic term) to get a differential equation for $\delta(t)$. Show that this leads to

$$\frac{\delta^2}{2\mathcal{Q}_{AB}} + \frac{\delta}{k_1'} = \bar{V}_B c_{A0} t \quad (19B.4-5)$$

which agrees with experimental data.¹ Interpret the result.

19.1-18:

$$\frac{\partial c_A}{\partial t} = \mathcal{D}_{AB} \nabla^2 c_A$$

Quasi-Stake: $\frac{\partial c_A}{\partial t} = 0$

BC:

$$c_A(z=0) = c_{A0}$$

$$c_A(z=\delta) = c_{A\delta}$$

$$\frac{dN_{A,z}}{dz} = 0, \quad \frac{dN_{B,z}}{dz} = 0, \quad \frac{dN_{C,z}}{dz} = 0$$

Stoichiometrically: $-N_{A,z} = 0$

$$\frac{dN_{B,z}}{dz} = \frac{dN_{C,z}}{dz}$$

$$N_{A,z} = C_1$$

$$c_A(z) = C_1 z + C_2$$

$$c_A(z=0) = c_{A0} = C_1(0) + C_2$$

$$C_2 = c_{A0}$$

$$c_A(z=\delta) = c_{A\delta} = C_1 \delta + c_{A0}$$

$$C_1 = \frac{c_{A\delta} - c_{A0}}{\delta}$$

$$c_A = \frac{z}{\delta} (c_{A\delta} - c_{A0}) + c_{A0}$$

$$c_A(z) = c_{A0} - (c_{A0} - c_{A\delta}) \frac{z}{\delta}$$

$$b.) \frac{dc_A}{dz} = -D_{AB} \frac{dc_A}{dz}$$

$$\frac{d}{dt} \left(\int_0^{\delta(t)} c_A(z) dz \right) = \int_0^{\delta(t)} \frac{\partial c_A(z, t)}{\partial t} dz + \left(f(\delta(t), t) \frac{d\delta(t)}{dt} - f(0, \delta(t)) \frac{\partial \delta}{\partial t} \right)$$

Rate of O_2 lost w/time

$$-R_A + c_{A\delta} \frac{\partial \delta}{\partial t} = -D_{AB} \frac{dc_A}{dz}$$

$$c_{A\delta} \frac{\partial \delta}{\partial t} = -D_{AB} \frac{dc_A}{dz} + R_A(c_{A\delta}(t), k_1')$$

$$c_{A\delta} \frac{\partial \delta}{\partial t} = -D_{AB} \frac{dc_A}{dz} - k_1' c_{A\delta}$$

$$c.) \frac{d}{dt} c_C = -D_{AC} \frac{dc_C}{dz} + R_C$$

$$c_C(\delta(t), z) = c_C \cdot \delta$$

$$c_C \frac{\partial \delta}{\partial t} = R_C \quad R_C = R_A$$

$$c_C = \frac{1}{V_C} \left(\frac{1}{V_C} \frac{\partial \delta}{\partial t} = k_1' c_{A\delta} \right)$$

$$d.) c_{A\delta} \frac{\partial \delta}{\partial t} = -D_{AB} \frac{\partial c_A}{\partial z} - k_1' c_{A\delta}$$

$$\frac{1}{V_B} \frac{\partial \delta}{\partial t} = k_1' c_{A\delta} \quad \frac{\partial c_A}{\partial z} = \frac{c_{A\delta} - c_{A0}}{\delta}$$

$$\frac{\partial \delta}{\partial t} = k_1' c_{A\delta} \bar{V}_B$$

$$c_{A\delta} k_1' c_{A\delta} \bar{V}_B = -D_{AB} \frac{c_{A\delta} - c_{A0}}{\delta} - k_1' c_{A\delta}$$

$$\frac{c_{A\delta} k_1' c_{A\delta} \bar{V}_B \delta + k_1' c_{A\delta} \delta}{D_{AB}} = c_{A\delta} - c_{A0}$$

$$- \frac{c_{A\delta}^2 \left(\frac{k_1' \bar{V}_B}{D_{AB}} \right)}{D_{AB}} - \frac{k_1' c_{A\delta} \delta}{D_{AB}} = c_{A\delta} - c_{A0}$$

$$c_{A\delta}^2 \left(\frac{k_1' \bar{V}_B}{D_{AB}} \right) + c_{A\delta} \left(1 + \frac{k_1' \delta}{D_{AB}} \right) = c_{A0}$$

19D.1. Derivation of the equation of continuity. In §19.1 the species equation of continuity is derived by making a mass balance on a small rectangular volume $\Delta x \Delta y \Delta z$ fixed in space.

(a) Repeat the derivation for an arbitrarily shaped volume element V with a sufficiently smooth fixed boundary S . Show that the species mass balance can be written as

$$\frac{d}{dt} \int_V \rho_A dV = - \int_S (\mathbf{n} \cdot \mathbf{n}_A) dS + \int_V r_A dV \quad (19D.1-1)$$

Use the Gauss divergence theorem to convert the surface integral to a volume integral, and then obtain Eq. 19.1-6.

(b) Repeat the derivation using a region of fluid contained within a surface, each point of which is moving with local mass average velocity.

flux of mass $= n_A$

$$\underbrace{\quad}_{m_{A,in} - m_{A,out}}$$

$$\text{Accumulation} = m_0 - m + r_A V$$

$$m_A = \rho_A V$$

$$dm_A = \rho_A dV$$

$$\int_0^{m_T} dm_A = \int_0^V \rho_A dV$$

$$m_T = \int_0^V \rho_A dV$$

$$\frac{d}{dt} \int_V \rho_A dV =$$

$$\frac{d}{dt} \int_V \rho_A dV$$

$$a.) \frac{d}{dt} \int_V \rho_A dV = - \int_S (\mathbf{n} \cdot \mathbf{n}_A) dS + \int_V r_A dV$$

$$\int_S (\mathbf{n} \cdot \mathbf{n}_A) dS = \int_V \nabla \cdot \mathbf{n}_A dV$$

$$\frac{d}{dt} \int_V \rho_A dV = - \int_V \nabla \cdot \mathbf{n}_A dV + \int_V r_A dV$$

$$\int_V \left[\frac{\partial \rho_A}{\partial t} + (\nabla \cdot \mathbf{n}_A) - r_A \right] dV = 0$$

$$\frac{\partial \rho_A}{\partial t} = - (\nabla \cdot \mathbf{n}_A) + r_A$$

equivalent

$$\mathbf{n}_A = \rho_A \mathbf{v} + \mathbf{j}_A$$

$$b.) \frac{d}{dt} \int_{V(t)} \rho_A dV = - \int_{S(t)} (\mathbf{n} \cdot \mathbf{j}_A) dS + \int_{V(t)} r_A dV$$

$$\int_{V(t)} \frac{\partial \rho_A}{\partial t} dV + \int_{S(t)} \rho_A (\mathbf{v} \cdot \mathbf{n}) dS = - \int_{S(t)} (\mathbf{n} \cdot \mathbf{j}_A) dS + \int_{V(t)} r_A dV$$

$$\int_{V(t)} \nabla \cdot (\rho_A \mathbf{v}) dV - \int_{V(t)} \nabla \cdot \mathbf{j}_A dV$$

$$\int_V \left[\frac{\partial \rho_A}{\partial t} + \nabla \cdot (\rho_A \mathbf{v}) + \nabla \cdot \mathbf{j}_A - r_A \right] dV = 0$$